

Julia Sisk-Reynes

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Expertise: using multiwavelength observations, computational methods, and theoretical modelling to investigate active galactic nuclei, the accretion flow onto massive black holes, and dark matter with X-ray imaging and spectroscopy and strong gravitational lensing with current and future instrumentation.

EMPLOYMENT AND RESEARCH

- Sep 2024 – **Postdoctoral Research Fellow**
Center for Astrophysics | Harvard & Smithsonian (CfA), Cambridge, MA
“Using gravitational lenses as high-resolution X-ray telescopes”.
PI: Dr. Daniel Schwartz.
- Jun 2025 – **Co-chair of the *NewAthena* Science Working Group 6 (SWG6) on Cosmology & Fundamental Physics**
Chair: Dr. Esra Bulbul (MPE, Garching, Germany). Co-chair: Dr. Florian Pacaud (University of Bonn, Germany).

EDUCATION

- Oct 2020 – Jul 2024 **Ph.D. in Astronomy**
Institute of Astronomy (IoA), University of Cambridge, UK
Thesis: “Unveiling fundamental physics with high-resolution X-ray spectroscopy of active galactic nuclei”.
Supervisors: Prof. Christopher Reynolds (University of Maryland, College Park, MD) and Dr. James Matthews (University of Oxford, UK).
- Sep 2016 – Jun 2020 **MPhys (BSc + Master’s) in Physics** (First Class Honours)
The University of Manchester, UK
Master’s thesis: “Building models for the thermal Sunyaev-Zel’dovich effect from analytic and numerical cluster pressure profiles”.
Supervisor: Prof. Scott Kay (University of Manchester, UK).

AWARDS AND PRIZES

- Summer 2023 **Flatiron Institute Summer School on Fluid Dynamics**
Attended a two-week Fluid Dynamics Summer School at the *Center for Computational Astrophysics, Flatiron Institute, New York, NY*. Attendance sponsored by the Simons Foundation.
- Jan 2022 **Paul Murdin Prize for the best published paper by a current student**
From IoA, Cambridge, UK, for [Sisk-Reynes et al., 2022, MNRAS, 510, 1](#)
- Oct 2020 – Mar 2024 **Full Studentship from the UK Science and Technologies Facilities Council**
To pursue a doctorate in astronomy at the Institute of Astronomy at the University of Cambridge, UK.
- Jun 2020 **Stellify Award from The University of Manchester**
Honorary recognition for extracurricular contributions made whilst an undergraduate student from 2016 – 2020.

PRESENTATIONS

May 2022 –	Thirty-seven presentations at international conferences, workshops, telescope collaborations, and group meetings. • Type: Invited (20), Contributed (17). • Format: Presential (29), Virtual (8). • Locations: US (16), UK (10), Europe (9), Americas (1), Asia (1). • Per year: 2026 (3), 2025 (11), 2024 (5), 2023 (9), 2022 (9). Selected recent invited presentations:
Aug 2026	Talk on behalf of <i>NewAthena</i> SWG6, E1.18, COSPAR, 2026, Florence, Italy
May 2026	Talk on lensed quasars, MIT Monday Afternoon Talks, MKI, Cambridge, MA
Apr 2026	Colloquium, UMBRELA Dialogues, CfA, Cambridge, MA
Dec 2025	Colloquium, Lin Group Meeting, Hong Kong University
Oct 2025	Talk on behalf of <i>NewAthena</i> SWG6, HEAD/AAS 22 Meeting, St Louis, MO
Oct 2025	Colloquium, Universidad Nacional Autónoma de México, Mexico
May 2025	Talk to the <i>AXIS</i> AGN SWG on an <i>AXIS</i> survey of massive black hole spins
Jan 2025	Colloquium, AGN group at NASA GSFC, Greenbelt, MD
Jun 2024	Talk to axion journal club, Oskar Klein Center, Stockholm University, Sweden
Apr 2024	Talk to the HEA Group, MPE, Garching, Germany

AWARDED TELESCOPE TIME

Jul 2024	Co-I , <i>Chandra</i> General Observer Program, Cycle 26 – 640 ks. PI: Dr. Helen Russell (University of Nottingham, UK). Target: Zw2089 cluster.
Jul 2024	Co-I , <i>Chandra</i> Legacy Program, Cycle 26 – 3 Ms. PI: Prof. Andrew Fabian (IoA, University of Cambridge, UK). Target: Perseus Cluster.

SELECTED ACADEMIC SERVICE

Telescope Allocation Time

Aug 2025	<i>XRISM</i> Guest Observer program reviewer, Cycle 2 (3 days, presential)
Mar – May 2025	<i>Chandra</i> Guest Observer program reviewer, Cycle 27 (4 days, virtual)
Nov 2024	<i>NICER</i> Guest Observer program reviewer, Cycle 7 (5 days, virtual)

Journal paper reviewer

Sept 2025 –	Physical Review D, Physical Review Letters
Sept 2025 –	Monthly Notices of the Royal Astronomical Society (MNRAS)

Invited speaker at discussion panels in conferences and meetings

Jun 2024	Dark matter in Astrophysical Laboratories workshop Kavli Institute for Cosmology (KICC), Cambridge, UK
Mar 2024	International Women's Day Discussion Panel for undergraduate students Women in Science Society, University of Cambridge, UK

Conference organization

Jul 2024	Local Organizing Committee, IoA 50-year anniversary, IoA, Cambridge, UK
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SKILLS

Programming	Bash, Python, tcl, Unix/Linux
Astronomy software	(Py)XSPEC, Sherpa, CIAO, ds9, SAOTrace
Numerical tools	ALPRO (axion code, advanced level), AREPO (basic level)
Languages	English, Catalan, Spanish

PRESS COVERAGE

May 2023	Athena Press Release on Sisk-Reynes et al., 2023, ApJ, 930, 1
Jun 2022	Chandra Press Release on Sisk-Reynes et al., 2022, MNRAS, 514, 2

STUDENT SUPERVISION

Sep 2025 –	Henry Hiscocks. Exchange master's student at CfA. Main supervisor.
Sep 2025 –	Charlotte Eades. Exchange master's student at CfA. Co-supervisor.
Nov 2024 –	Alex Ostridge. Exchange master's student at CfA. Advisor.
Jun – Aug 2022	Esin Gulhabar. Summer research student at IoA, Cambridge. Advisor.

Research areas: axions, black hole spins, machine learning, X-ray astrometry of lensed quasars.

TEACHING AND STUDENT SUPPORT

Oct 2022 – Jun 2023	Mathematics I Supervisor. Emmanuel College, University of Cambridge.
Jan – May 2021	Astrophysical Fluid Dynamics. Trinity College, University of Cambridge.
Sep 2017 – May 2019	Peer Assisted Study Sessions (PASS) Leader, University of Manchester.

SELECTED OUTREACH ACTIVITIES

Jul 2025	Speaker, Spanish Public Observatory (100 attendees), CfA, Cambridge, MA
Feb 2025	Speaker on visit to a bilingual school (100 attendees), Chelsea, MA
Mar 2024	Speaker, Lindemann Trust Science Day (120 attendees), Cambridge, UK
Mar 2023	Speaker, Lindemann Trust Science Day (100 attendees), Cambridge, UK

OTHER RELEVANT TRAINING

Sep 2025	Spins Workshop, Wake Forest University (2 days), Winston-Salem, NC
Feb 2025	3 rd XRISM Community Workshop (3 days), College Park, MD
Jul – Aug 2019	Summer research on direct dark matter searches, Manchester, UK

SELECTED INSTITUTIONAL SERVICE

At the Center for Astrophysics | Harvard & Smithsonian, Cambridge, MA

Apr 2025 –	Lead organizer, monthly collaborative discussions, cosmology journal club
Nov 2024 –	Organizer, weekly High Energy Astrophysics Division seminar series

At the Institute of Astronomy (IoA), University of Cambridge, UK

Oct 2021 – Apr 2024	Lead organizer, IoA International Women's Day celebration Responsibilities: keynote speakers, discussion panel (attendees: 100/year).
Oct 2021 – Apr 2024	Lead organizer, Newnham College MCR Seminar Speaker Series Organized a 4-lecture series on the legacies of slavery (attendees: 70).
Jan 2021 – Oct 2023	Representative at events for the general public and prospective applicants
Oct 2022 – Jun 2023	Representative, Computer Users Committee
Dec 2020 – Mar 2024	Representative for 2020 entry Ph.D. students

At the University of Manchester, UK

Oct 2019 – Jun 2020	Class representative for final-year Physics students
Mar 2017 – Feb 2020	Regular volunteer at events organized by the Physics Outreach Society
Mar 2017 – Jun 2020	Lead committee member, undergraduate Nuclear & Particle Physics Society Organized monthly seminars and a 4-day trip to CERN (50 attendees).

COLLABORATIONS AND SCIENTIFIC MEMBERSHIP

May 2025 –	Co-chair, <i>NewAthena</i> SWG6 - Cosmology and Fundamental Physics
May 2025 –	Member, <i>NewAthena</i> SWG2 - Galaxies and Supermassive Black Holes
Dec 2024 –	Member, <i>AXIS</i> Working Group on Active Galactic Nuclei
May 2023 – Oct 2024	Member, <i>STROBE-X</i> Working Group on Cosmology
Jun 2025 –	International Astronomical Union (IAU)
Sep 2024 –	American Astronomical Society (AAS)
Sep 2024 –	AAS High Energy Astrophysics Division (HEAD)
Sep 2021 –	European Astronomical Society (EAS)
Dec 2021 – Jun 2024	Royal Astronomical Society (RAS)

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Publication List

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Authored or co-authored eleven publications which are either published or in review, including an invited review article. As of February 2026, these have been cited 216 times in the scientific literature (h-index: 5). Publications by mentored students appear with a dagger (†). Accepted and published refereed journal papers appear with an asterisk (*).

FIRST-AUTHOR PUBLICATIONS

1. **Determining the spatial origin of X-ray and optical emission in the $z=3.1$ strongly lensed radio-quiet quasar GraL J065904.1+162909 to hundreds of parsecs.*
Sisk-Reynes J., Schwartz D., Barnacka A., Spingola C. & Migliori G.
Accepted for publication in ApJ in Jan 2026 (first submitted version: [arXiv: 2509.22797](#)).
2. *The detergent particle.*
Sisk-Reynes J.
Invited review article.
Published Dec 2024; [RAS Astronomy & Geophysics, December 2024, Volume 65, Issue 6](#).
3. **Physics Beyond the Standard Model with Future X-ray Observatories: Projected Constraints on Very-Light Axion-Like Particles with Athena and AXIS.*
Sisk-Reynes J., Reynolds C., Parker M., Matthews J. & Marsh M. C. D.
Published Jun 2023; [ApJ, 951, 1](#).
4. **Evidence for a moderate spin from X-ray reflection of the high-mass supermassive black hole in the cluster-hosted quasar H1821+643.*
Sisk-Reynes J., Reynolds C., Matthews J. & Smith R.
Published Jul 2022; [MNRAS, 514, 2](#).
5. **New constraints on light axion-like particles using Chandra transmission grating spectroscopy of the powerful cluster-hosted quasar H1821+643* (citations: 110)
Sisk-Reynes J., Matthews J., Reynolds C., Russell H., Smith R. & Marsh M. C. D.
Published Feb 2022; [MNRAS, 510, 1](#).

CO-AUTHOR PUBLICATIONS

6. †*Speeding up Gravitational Lens Mass Models with Machine Learning: Applications in X-ray Astronomy.*
Ostridge A., Martinez-Galarza R., Schwartz D., **Sisk-Reynes J.** & Barnacka A.
In review after submission to MNRAS in Jul 2025.
7. **Getting More Out of Black Hole Superradiance: a Statistically Rigorous Approach to Ultralight Boson Constraints from Black Hole Spin Measurements.*
Hoof S., Marsh D. J. E., **Sisk-Reynes J.**, Matthews J. & Reynolds C.
Published in Feb 2026; [MNRAS, 546, 2](#).
8. **How Do Magnetic Field Models Affect Astrophysical Limits on Light Axion-like Particles? An X-ray Case Study with NGC 1275.*
Matthews J., Marsh M. C. D., Reynolds C., **Sisk-Reynes J.** & Rodman P.
Published May 2022; [ApJ, 930, 1](#).

CO-AUTHOR REFEREED CONFERENCE PROCEEDINGS

9. **Evidence for supermassive black hole binaries.*
Krause M., Bourne M., Britzen S., Foord A., Greene J. E., Habouzit M., Horton M. A., Mayer L., Middleton H., Nealon R., **Sisk-Reynes J.**, Reynolds C. & Sijacki D.
Published Nov 2025; [PASA, 42, ID162](#).
Proceedings of a RAS Specialist Discussion Meeting in Apr 2023.
10. **Current and Future constraints on Very-Light Axion-Like Particles from X-ray observations of cluster-hosted AGN.*
Sisk-Reynes J., Reynolds C. & Matthews, J.
Published in Mar 2023; [Mem. S.A.It. Vol. 94, 29](#).
From Dark Matter to Machine Learning: A collection of proceedings from the S3 and S11 symposia at the European Astronomical Society's 2022 annual meeting in Valencia, Spain.

CO-AUTHOR REFEREED SCIENCE BOOKS FOR FUTURE MISSIONS

11. *The Advanced X-ray Imaging Satellite Community Science Book.*
Koss M., [...], **Sisk-Reynes J.**, [...].
Released in Oct 2025; [arXiv: 2511.00253](#).
Contributed by leading two proposals (one on physics beyond the Standard Model and one on massive black hole spins) for the *AXIS* Guest Observer program.

MANUSCRIPTS TO BE SUBMITTED BY MARCH 2026

12. *Spin demographics of active supermassive black holes: Updated measurements from X-ray reflection and future opportunities.*
Sisk-Reynes J., [...].
To be submitted to *Galaxies* as an invited contribution to the [Physics of Black Hole Spins issue](#).
13. *A new Chandra X-ray spectrum of the $z = 3.273$ lensed dual AGN candidate MG B2016+112.*
Schwartz D., **Sisk-Reynes J.**, [...].
To be submitted to ApJL.